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REVIEW ARTICLE

Analyzing Activity and Injury: Lessons Learned from the Acute:Chronic Workload Ratio

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Abstract

Injuries occur when an athlete performs a greater amount of activity than what their body can withstand. To maximize the positive effects of training while avoiding injuries, athletes and coaches need to determine safe activity levels. The International Olympic Committee has recommended using the acute:chronic workload ratio (ACWR) to monitor injury risk and has provided thresholds to minimize risk when designing training programs. However, there are several limitations to the ACWR and how it has been analyzed which impact the validity of current recommendations and should discourage its use. This review aims to discuss previously published and novel challenges with the ACWR, and strategies to improve current analytical methods. In the first part of this review, we discuss challenges inherent to the ACWR. We explain why using a ratio to represent changes in activity may not always be appropriate. We also show that using exponentially weighted moving averages to calculate the ACWR results in an initial load problem, and discuss their inapplicability to sports where athletes taper their activity. In the second part, we discuss challenges with how the ACWR has been implemented. We cover problems with discretization, sparse data, bias in injured athletes, unmeasured and time-varying confounding, and application to subsequent injuries. In the third part, conditional on well-conceived study design, we discuss alternative causal-inference based analytical strategies that may avoid major flaws in studies on changes in activity and injury occurrence.

1 Introduction

Injuries are an unintended consequence of activity and a major concern for elite and recreational athletes. Injuries may result in morbidity and time lost from activity [1], and may even terminate an elite athlete's career or a recreational athlete's desire to be active. Quantifying the relationship between physical activity and injury risk to determine a safe level of activity should improve the well-being of all athletes.

When an individual performs activity, their muscles, tendons and bones are exposed to physical stress. While these tissues may adapt and increase their strength [2], large increases in physical stress that surpass their strength will cause damage and injury. Although some aspects of physical stress can be measured (e.g. distance travelled), we will never be able to measure all the physical stressors on an individual. Yet, if most of the relevant stresses are measured, we can perform analyses that will still provide useful guidance. In this article, we collectively refer to all stresses during activity that might cause injury as load.

An unwritten rule of thumb for many years was that injury would be minimized if activity were increased by less than 10–20% per week [3–5]. Recently, there has been increased interest in using changes in physical activity patterns to predict injury risk. Hulin et al. introduced the concept of an acute:chronic workload ratio (ACWR) to measure changes in activity [6]. The ACWR is calculated as the acute (i.e. recent) load divided by the chronic (i.e. long term) load [6]. Although the acute load is generally defined as the activity performed in the past week and the chronic load as a weekly average of activity in the past 4 weeks, the choice of

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Key Points

The calculation of the acute:chronic workload ratio is recommended by the International Olympic Committee to guide training and minimize injury risk. However, it has several conceptual and mathematical limitations that should discourage its use.

The acute:chronic workload ratio may not scale accurately and may not be the most meaningful measure of load.

The exponentially weighted moving average suffers from an initial load problem and is not applicable to all sports.

Alternative causal inference-based strategies should be adopted to analyze the effects of changing activity on injury.

time windows may differ depending on the sport and training schedule [7], and more research is needed to determine the most appropriate windows. The unit of load depends on the context; it may be quantified using ratings of perceived exertion, or measures related to the work performed (e.g. distance travelled, balls bowled, weight lifted) [8].

In studies of cricket, rugby, Australian Football, and soccer players, increases in ACWR were associated with increases in injury risk [6, 9–12]. A general model for the relationship between one's ACWR and their risk of injury was generated using data from cricket, rugby, and Australian Football players (Fig. 1) [13]. It identified ratios in the "sweet spot" between 0.8 and 1.3 as being associated with the lowest risk of injury. Injury risks increased with decreasing ratios below 0.8, and increasing ratios above 1.3 [14]. These results supported the rule of thumb for a 10–20% increase in load per week. Although the ACWR was developed as a tool to predict injury, an International Olympic Committee (IOC) consensus statement uses the general model as a guideline for training practices [14], which requires a causal interpretation (i.e. changing load so the ACWR is greater than 1.3 will lead to more injuries than changing load so the ACWR is between 0.8 and 1.3).

Although many studies have since used the ACWR to predict injury risk, some authors have noted limitations with it. These include feasibility challenges, wherein load cannot always be measured accurately and frequently [15, 16]. For instance, loads are often measured using measures (e.g. balls bowled, distance run, session rating of perceived exertion) that may have complicated non-linear relationships with the load on one's tissues that cause injury. Further, if different load measures (e.g. balls bowled, distance run) are not interchangeable, results will be heterogeneous and recommendations based on one measure will not be applicable to all [17]. These challenges are not unique to the ACWR. For example, pain might only be measured once per week and may be measured using many different scales. However, even if loads were measured accurately and frequently, there are major methodological flaws in the ACWR that should invalidate its use. Similarly, the choice of broad injury definitions (e.g. all injuries) versus narrow definition of injuries (e.g. lower extremity tendinopathies lasting at least 7 days) needs to be considered when applying recommendations to particular clinical contexts.

The purpose of this article is to demonstrate that the ACWR and associated models are seriously flawed,

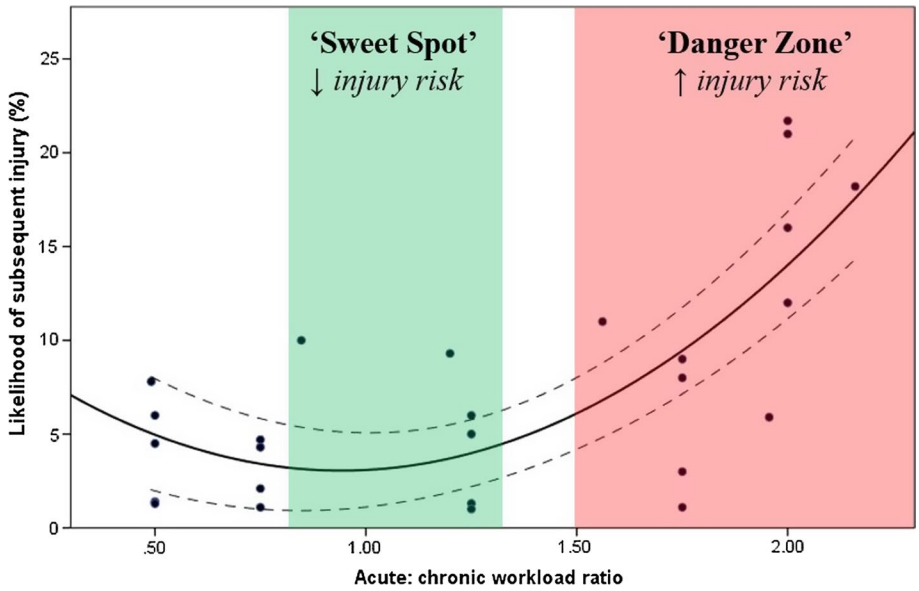

Fig. 1 Established relationship between the acute:chronic workload ratio and injury risk based on studies from three different sports. The green-shaded area covers acute:chronic workload ratios within a range of approximately 0.8–1.3, and represents a 'sweet spot' where injury risk is low. The red-shaded area covers acute:chronic workload ratios of 1.5 and above, and represents a 'danger zone' where injury risk is high (reproduced from Gabbett [8], with permission)

especially if interpreted causally and used to design training programs and to discuss more appropriate strategies. In Sect. 2, we discuss the inherent limitations associated with the ACWR. In Sect. 3, we discuss limitations with how the ACWR has been used to develop activity-injury risk models and what we can learn from them in general when analyzing activity and injury risk. In Sect. 4, we provide novel analytical strategies for quantifying the relationship between changes in activity and injury risk, and a summary of the type of data that would be necessary to avoid some of the more serious limitations presented by current strategies.

2 Limitations of the ACWR

In this section, we discuss inherent limitations of the ACWR that make it a flawed measure of changes in activity over time.

2.1 The Coupled ACWR is a Measure of the Proportion of Activity Rather than the Change

The ACWR was intended as a preventive tool to indicate whether an athlete has sufficiently prepared for their upcoming (acute) load by normalising it to their previous (chronic) load [9]. If an athlete has a low acute load relative to their chronic load, they are decreasing their relative activity and will have a low ACWR. Athletes with low ACWRs are performing within the limits of what they have prepared for (i.e. loads that they had already been doing without injury), and are, therefore, expected to have lower risks of injury. Conversely, if an athlete has a high acute load relative to their chronic load, they will have a high ACWR. These athletes are expected to exceed the level of training they have prepared for, resulting in an increased risk of injury [8]. Athletes with very low ACWRs also have an apparent increased risk of injury (Fig. 1). This finding may be due to bias, as explained in Sect. 3.3.

Although the ACWR acts as a marker of athlete preparedness based on changes in activity [8], the conventional measure is actually a proportion (i.e. the amount that a part represents in relation to the whole) rather than a true measure of change. The denominator of the ACWR (chronic load) typically includes 4 weeks of load. The numerator (acute load) typically includes the most recent week, representing one part of the denominator. As activity performed as part of the acute load is included in the calculation of the chronic load, this form of the ACWR is a "mathematically coupled" measure [18] that often underestimates change. The use of an "uncoupled" measure, where the acute load is excluded from the chronic load calculation, has also been proposed [18]. In this section, we will discuss limitations associated with the coupled measure, and how they might be resolved with the uncoupled measure.

There are two important limitations to the conventional coupled approach. First, it creates a spurious correlation between acute and chronic loads [19]. This creates an apparent decrease in the variability of load between athletes, potentially leading to inappropriate inferences [19]. Using an uncoupled measure removes the correlation and increases the between-athlete variability [19].

Second, the coupled measure creates a theoretical maximum ACWR of 4, regardless of the magnitude of the absolute change in load [18]. For example, consider an individual who has planned activity for the upcoming week, but was inactive in the 33 weeks prior. The numerator for their ACWR will be the load in that week (L1). The denominator will be the average of the total load from the 4 weeks, which is $0+0+0+L1$ (i.e. $L1/4$). The ACWR is, therefore,

$$\frac{L1}{\left(\frac{L1}{4}\right)} = 4,$$

no matter the magnitude of L1. Therefore, the coupled ACWR is unlikely to hold as a good predictor of injury risk at relatively high acute loads. The uncoupled measure excludes the acute load week from the chronic load calculation and has no such maximum bound [18]. Therefore, one might expect it to better differentiate changes in the load that have different absolute magnitudes.

2.2 The Uncoupled ACWR May Not Be the Most Meaningful Measure of Load

The uncoupled ACWR solves the above limitations of the coupled measure and follows the same patterns as the conventional calculation where large increases in acute load relative to the chronic load are associated with increased injury risk [20]. However, it still requires strong assumptions. The uncoupled ACWR represents changes in activity by combining loads at different times into a simple ratio. Using the uncoupled ACWR to model injury risk assumes that a simple ratio of acute and chronic loads is the best predictor of injury. An alternative strategy without this assumption would be to model acute and chronic loads as separate variables. The ratio method also makes it difficult to determine whether any observed relationship between the ACWR and injury is due to the ratio itself, the chronic load denominator, or both.

In other areas of sports medicine, we have learned that simple ratios are often not the optimal method (e.g. body mass index, kg/m^2). The reviewers of our manuscript also highlighted some papers that suggest normalizing the acute load to the chronic load through a simple ratio will lead to inaccurate scaling across the range of loads unless the

relationship between the numerator and denominator is linear with an intercept through the origin [21, 22]. The uncoupled acute and chronic loads were only trivially correlated in a small sample of 24 English Premier League players monitored over a complete season [23]. Statistically, when and how to normalize data is complicated and a full discussion is beyond the scope of this paper. Our alternative strategies avoid the challenges of using a simple ratio measure to express changes in activity over time (Sect. 4). Nevertheless, researchers should always examine the underlying assumptions that underpin any particular mathematical formulation before considering any resultant variable as a valid exposure in a causal or predictive statistical model.

2.3 The ACWR Uses Unweighted Averages to Calculate Load

In the original formulation of the ACWR, the chronic load is defined as the weekly average of the load performed in the past 4 weeks [6]. Each of the 4 weeks is weighted equally in the calculation. While straightforward to calculate, the equal weighting obscures weekly variations in load and neglects that the effects of training decay over time [24]. Menaspà illustrated how 3 different athletes with vastly different training patterns can have the same ACWR (Fig. 2) [25]. Although each athlete would be assigned the same injury risk based on their shared ACWR, their true injury risks would conceivably differ depending on how their load was spread over the 4 weeks.

Similar problems arise with using a single weekly value for the acute load. Two athletes with the same acute load may have different daily variations in the activity that affect their risk of injury. For example, an athlete who performs all of his or her activity on a single day might be at higher risk of injury than an athlete who performs the same amount of activity spread over 7 days.

The underlying issue is that an injury is typically more likely to be associated with one's most recent load than load performed weeks prior. To solve this problem, some authors suggested that average loads be calculated by providing more weight to recent loads and less weight to loads in the distant past [25, 26]. Williams et al. proposed a weighting method that uses exponentially weighted moving averages (EWMA) for the calculation of chronic and acute loads [26]. The EWMA applies exponentially decreasing weights for loads performed on each prior day, such that loads performed on the first day of the chronic load calculation (i.e. the day furthest away from the day of calculation) and the first day of the acute load calculation should theoretically be weighted the least. Therefore, those who engage in greater recent activity will have higher loads, which may affect whether they are in a 'sweet spot' for minimizing injury risk.

Using EWMA weighting may better reflect changes in activity patterns and their effects on the injury. In one small study, R^2 increased from 0.21 in the traditional unweighted

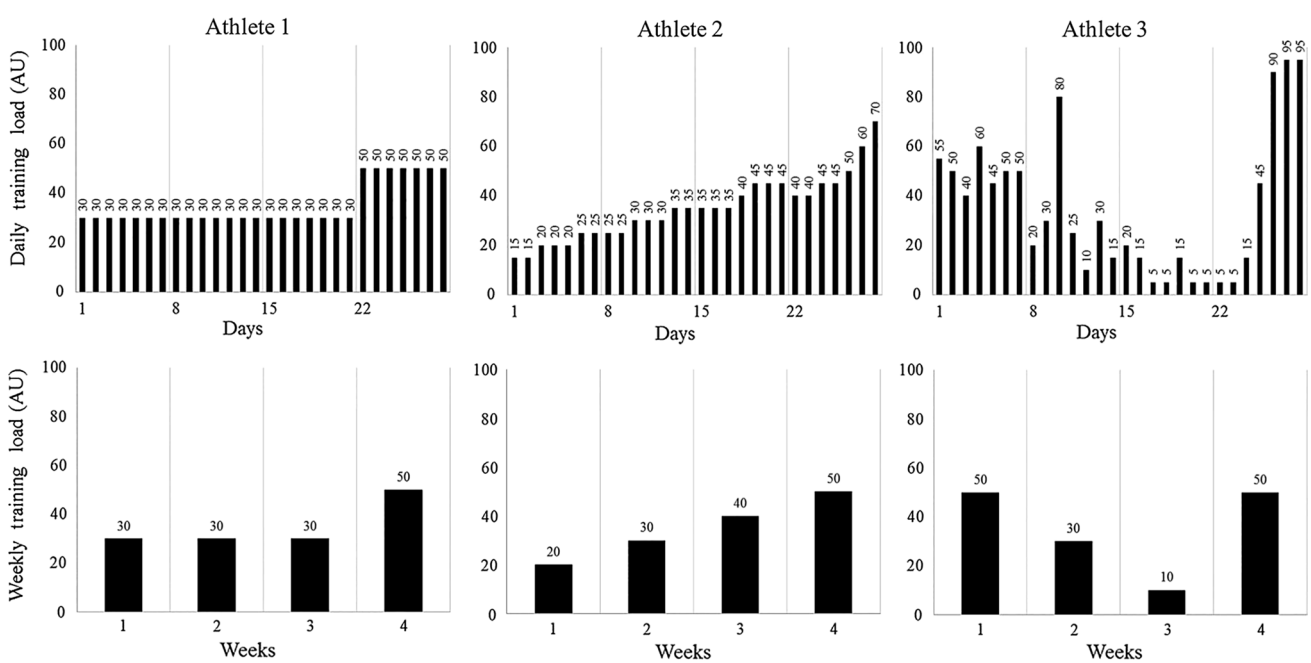

Fig. 2 The daily training loads of three athletes are shown at the top and their weekly training loads are shown at the bottom. Each athlete has the same acute load (50 arbitrary units), chronic load (35 arbitrary units), and acute:chronic workload ratio (1.43) despite having different training schedules (reproduced from Menaspà [25], with permission)

method to 0.87 using weighted averages [27]. While this may provide evidence that EWMA may be a more sensitive predictor of injury, the authors used conventional logistic regression without accounting for repeated measures (further discussed in Sect. 3.2). Yet, despite the purpose of the EWMA approach to address the shortcomings of conventional rolling averages [25], there are several important limitations to this method that have not yet been reported.

The purpose of EWMA is to give more recent activity more weight, and activity further in the past less weight. However, mathematically, the initial load on Day 0 that is needed to calculate EWMA-weighted ACWRs may take on weight that is much higher than it should be, and in some cases even larger than the most recent load. The EWMA calculation uses decay constants to apply decreasing weights to activity further in the past. These decay constants are arbitrary, as the true decay rates of the effects of load are unknown, and would likely differ between individuals and sporting contexts. When the decay constant is less than 0.5, the value at Day 0 (most distant day from the day of calculation) takes on a greater weight than at least the next day, and sometimes many subsequent days (Electronic Supplementary Material Appendix S1). The difference between the weights at Day 0 and Day 1 increases as the decay constant gets closer to zero (Electronic Supplementary Material Table S1). This problem occurs with the proposed decay constants by Williams et al. when used to calculate loads for 28 days of activity. They proposed constants of 0.25 for the acute load and 0.068 for the chronic load [26], regardless of the individual and sporting context. Although the initial load on Day 0 takes on a larger weight than subsequent days for both acute and chronic calculations, the decay constant is relatively large for the acute load, resulting in a fast rate of decay and a negligible difference between weights. The chronic load decay constant is much closer to zero, resulting in a large contribution from the load on Day 0 towards the weighted average. In fact, the contribution of the load on Day 0 is 1.9 times the contribution of the most recent load (Day 28) to the weighted average, even though the most recent load should contribute the most weight (Fig. 3).

This problem occurs because a value must be entered for the "initial load" on Day 0 to calculate the ACWR on Day 1, even though the load on Day 0 is not observed. One may choose to assign the first recorded load to the initial load [27], or use an arbitrary value, with 0 being an extreme case. Either value may lead to the initial load taking on an inappropriately large weight in the EWMA. However, with enough days included in the calculation, the EWMAs for an athlete with a positive initial load and with an initial load of 0 will converge. We illustrate this concept with an example using training loads from athlete 3 in Menaspà's paper [25]. We compare two scenarios—one where both the initial load and load on Day 1 are 55, which is the first recorded load for athlete 3, and one where the initial load is 0 and the load on Day 1 is 55. If this athlete repeats the same training profile 3 times consecutively, the two ACWRs will converge after around 50 days (Fig. 4). This means that at least 50 days of activity are needed to reach convergence and avoid the initial load problem. In other words, we would have to ignore the first 50 days of information to obtain a valid estimate of injury risk. Further, this would only be valid for people who did not get injured during the first 50 days of training. Therefore, the methods of calculation for the decay constant or the EWMA itself would need to be changed.

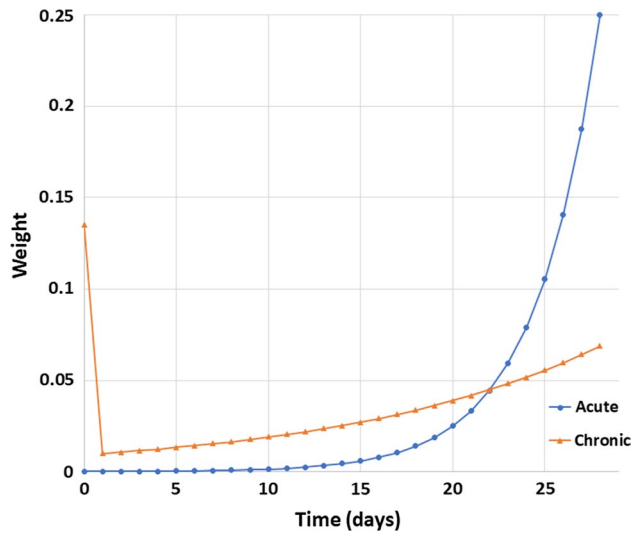

Fig. 3 Weights for acute and chronic loads using the exponentially weighted moving average with 28 time points. The initial load takes on an inappropriately large weight for the chronic load, but its weight is negligible for the acute load

2.4 The ACWR Does Not Account for Tapering Before Competitions

The EWMA attempts to account for the belief that activity performed in the distant past is less likely than recent activity to cause injury. Yet, this is not necessarily true in certain contexts. Previous studies that demonstrated the utility of the EWMA looked at sports with competitive seasons [27]. However, when athletes train for a marathon, they typically taper their training load immediately prior to a race to recover from prior heavy training [28]. Tapering aims to reduce fatigue and increase recovery from training stress [29], thereby reducing the risk of injury [30]. Similar training techniques are used by athletes who partake in sports with competitions lasting a short period of time, such as swimming, cycling, triathlon, and strength training [29]. In this context, tapering activity the few weeks before the competition decreases the chronic load and thereby increases the ACWR during the competition. If rest prior to increased

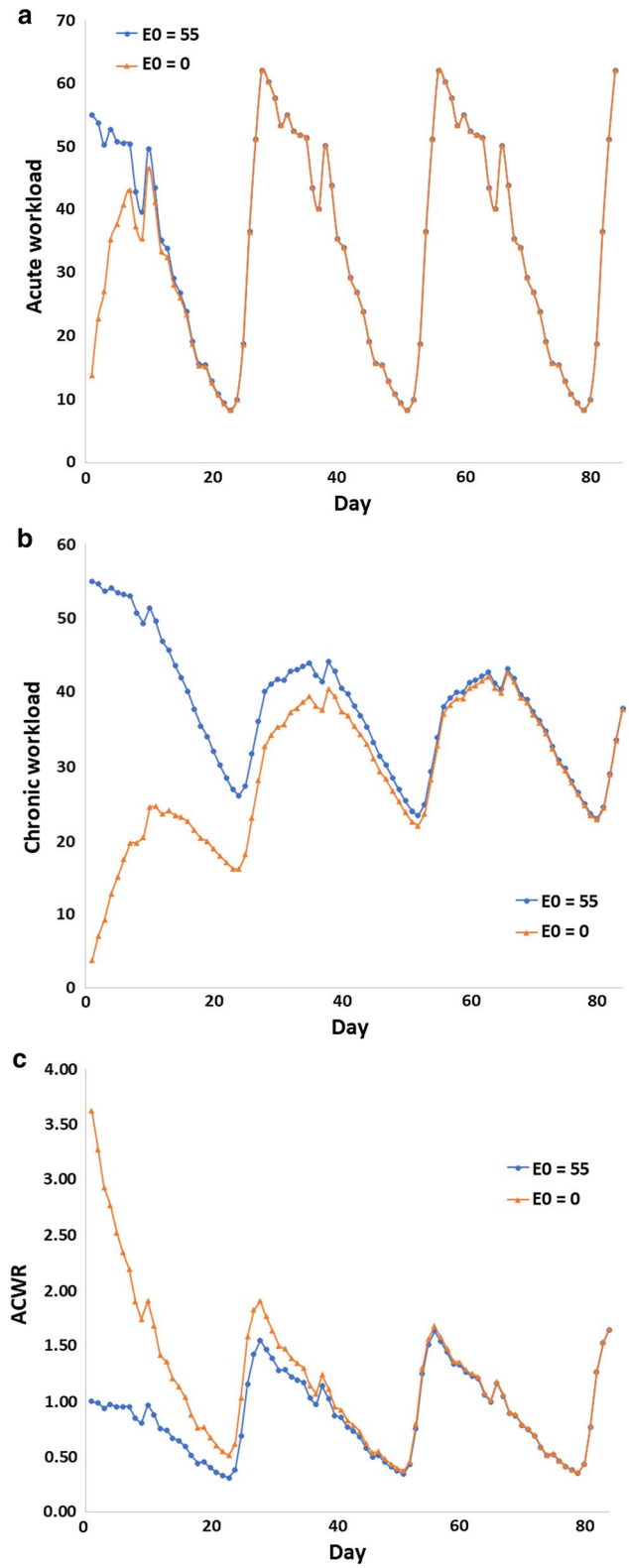

Fig. 4 Acute loads (**a**), chronic loads (**b**), and the acute:chronic workload ratio (ACWR) (**c**) calculated using the exponentially weighted moving average (EWMA) for two different initial loads (E0). The training schedule begins with a load of 55 units on Day 1. The blue line uses 0 units for the initial load (the unobserved value for load experienced prior to the first day of training used to calculate load on the first day; E0 = 0), and the orange line uses 55 units for the initial load (E0 = 55). The ACWR on the first day of training (Day 1) is 3.65 when the initial load is 0, and 1.00 when the initial load is 55. The two ratios converge after 50 days

activity during competition reduces injury risk as expected, the ACWR will be a poor predictor of risk. Therefore, a different model and set of recommendations must be used for these contexts.

3 Limitations of How the ACWR Has Been Implemented

In Sect. 2, we discussed fundamental limitations of the ACWR that make it a flawed measure of changes in activity over time. In this section, we discuss limitations with how the ACWR has been implemented rather than a limitation of the ACWR concept itself. These limitations are applicable to the modelling of the ACWR as well as valid load measures, and may greatly impact the validity of study conclusions.

3.1 The ACWR is a Continuous Measure That Has Been Discretized

The ACWR can take on different values depending on how load is measured. Loads can be quantified in a variety of ways, including distance travelled, weight lifted, and session-rating of perceived exertion [6]. Regardless of the method of measurement and whether it is truly continuous, the ACWR is essentially treated as a continuous measure with units that can conceivably take on any positive value in the uncoupled measure or any positive value up to 4 in the coupled measure.

The majority of studies utilizing the ACWR as a predictor of injury risk have categorized loads and ratios into bins rather than treating the ACWR as a continuous measure [6, 7, 13]. There are several apparent benefits of the categorization of loads into discrete ranges. Athletes, coaches, and clinicians find categorizations helpful in formulating a training plan. Rather than calculating an exact amount of activity to perform to minimize injury, athletes can determine whether they can safely engage in low, moderate, or high amounts of activity based on their chronic load, offering them more flexibility in their training and easing interpretability. Discretization of a continuous measure such as the ACWR can also increase efficiency and interpretability when developing models [31]. However, despite the apparent simplicity and practicality of discretization, it often creates problems in analyses and should be cautioned against.

Discretization prior to model selection in an analysis, as is typically done with the ACWR, results in a loss of

information and may hide the true relationship between variables [32]. Furthermore, different studies have binned ACWRs differently and used different reference levels [6, 9, 33, 34], making it difficult to extrapolate findings and create a generalized model for injury risk. In simulations of training load and injury risk, discretization of the ACWR resulted in increased false discovery rates and less accurate estimates when compared to continuous models [35].

Analyses are particularly flawed when discretized measures are modelled as continuous, as is the case with the established ACWR-injury risk model (Fig. 1). Despite each point corresponding to the mid- or endpoint of a discrete range of ACWR values, the data were inappropriately analyzed as continuous and fitted with a polynomial line [13]. Given that the data are not continuous, and that the points used in the analyses do not necessarily correspond to the measured likelihood of injury at different ACWRs, it is unlikely that the model represents the true relationship between physical activity and injury risk [17].

While we focus on the limitations of how the ACWR has been analyzed, discretization before modelling creates limitations for any measure of activity. A more appropriate analysis that minimizes the potential loss of data and bias is to use continuous data to develop injury risk models. To help athletes and coaches interpret the results, thresholds for the exposure (e.g. acute load) could be provided that correspond to different risk levels (e.g. low, moderate, high), based on the continuous model.

3.2 The Established ACWR Model is Based on Sparse Evidence from Under-Powered Studies

Sparse data problems are present in both original research data and evidence synthesis across studies. Studies that have used the ACWR to predict injury risk have been limited by small sample sizes, likely due to the difficulties of following a large group of participants over a long enough time to get a sufficient number of injuries. Nielsen et al. noted that out of 35 studies examining the relationship between loads and injury risk, only 11 had a sample size greater than 150 [36]. While an adequate sample size is necessary to establish an accurate activity-injury risk model [37], it is uncommon for studies to include a required sample size calculation. Researchers should justify their choice of sample size for full and transparent reporting [38–40]. Approaches to calculate required sample sizes are beyond the scope of this article. Readers interested in prediction models specifically may find the recent article by Riley et al. helpful [37].

In original research data, use of discretization as described above can amplify sparse data challenges. When the exposure is a categorical variable (e.g. discretized ACWR) and the outcome is non-linear (e.g. dichotomous outcome such as injury), one generally requires 5 events for each level of each combination of covariates to avoid bias [41]. For example, categorizing the ACWR into low-, medium-, and high-risk creates 3 levels. In a univariate analysis, one would need at least 15 events (i.e. injuries), 5 of which should occur in the low-risk group. This requirement is independent of the number of participants in each group. If one adjusts for additional variables like sex and age, 5 events would be required in each of the level combinations (e.g. 5 events in females aged 20–25 with a low ACWR, 5 events in males aged 20–25 with a low ACWR, etc.). If the exposure is continuous, one could instead fit splines, minimizing the challenges associated with sparse data [32].

The established model used in the IOC consensus statement (Fig. 1) is an important example of not having sufficient data for evidence synthesis. The model is based on data from three studies with sample sizes of 28, 53, and an unreported number of athletes [6, 9]. The number of injuries was reported for only one of these studies [9]. Furthermore, researchers used conventional logistic regression in their analyses [6, 9], which assumes each outcome (injury) is an independent event. Because repeated measures are not accounted for, results are overly precise and potentially biased [42].

ACWRs are discretized in the established model, likely due to the relative lack of data at each value. This is probably especially true for ACWRs at the end-points (0–0.5 or 2.0+), which were discretized as 0.5 or 2.0 respectively [13, 14], as the number of exposures to high ACWRs is likely to be very small. According to the model, the sweet spot for minimal injury risk is between 0.8 and 1.3, with a "danger zone" upwards of 1.5 [8]. However, the increase in injury risk appears to be driven completely by points at ACWR = 2.0, which also differ substantially from each other in their corresponding likelihood of injury. Limiting the data to ACWRs lower than 2.0 removes any apparent relationship between ACWR and injury risk.

Although this could indicate a "threshold" effect was the model valid, the discretization of endpoint data makes it impossible to determine whether there is a threshold, and if so, whether the increase in risk occurs at ACWRs of 2.0 or at higher values. Large sample sizes that avoid sparse data may provide more accurate training recommendations when modelling activity and injury risk. Researchers should make the appropriate calculations and be transparent when selecting a sample size, and recommendations should clearly outline the limitations due to sparse data if they are being disseminated. Any conclusions based on studies that have sparse data and use discretization should be interpreted with caution.

3.3 Increased Injury Risks at Low ACWR are Expected Due to Bias

The established ACWR-injury risk model shows an increasing risk of injury at ACWRs below 0.8 (Fig. 1). This suggests that performing less activity than one has prepared for in the previous weeks results in a higher risk of injury than performing the same level of activity. It also suggests that progressively lower amounts of activity result in further increases in injury risk. However, there is no biological theory that would support this hypothesis, as injury only occurs when the load exceeds the load capacity of the tissue [43]. Although this finding may be due to sparse data bias as mentioned above, the observed increase in injury risk associated with low ACWRs in Fig. 1 can also be explained by bias due to its method of calculation.

Activity data has typically been categorized into weekly blocks from Monday through Sunday based on calendar time [6, 9, 10, 12]. Consider the following example: two athletes have the same chronic load. One athlete has planned for 2 h of activity each day for the current week, while the other athlete has planned for 1 h of activity each day. If the larger load causes injury for Athlete 1 on Tuesday, they will have an acute load of 4 h. Meanwhile, Athlete 2 completes the week without injury, with an acute load of 7 h. Athlete 1 will have a lower ACWR than Athlete 2, despite performing more activity just prior to the injury. Because athletes who get injured earlier in the week will have systematically lower acute loads than athletes who complete the week without injury, the analytical strategies used for the ACWR will show an increased injury risk for low ratios even though there is no causal relationship.

There are three alternative methods of calculation that researchers have used in the literature. First, one could measure injury risk in the week subsequent to the acute load [9]. Although measuring injury risk for the subsequent week circumvents low ACWRs being associated with high risks of injuries from athletes getting injured early in the week, it ignores variations in the current week that likely impact injury risk. For example, an athlete with a low ACWR may have a sudden spike in activity in the subsequent week that causes injury. However, analyses would consider their low ACWR from the previous weeks to be predictive of injury, rather than their spike in recent activity.

Second, one could measure the ACWR of the current week with a two-week average load from the current and prior week [9]. Calculating the acute load as a two-week average faces the same problems as measuring injury risk in the current week—athletes who get injured early in the week will have a systematically lower ACWR, although to a lesser extent.

Third, some have calculated acute and chronic loads as daily moving averages, where the acute load is the average daily load in the last *X* days (e.g. 7 days) and the chronic load is the average daily load in the last *Y* days (e.g. 28 days) [7]. The ACWR for a particular day is the acute load divided by the chronic load, whereby the acute load excludes activity performed on that day. Therefore, injury risk is measured for the day subsequent to the last day included in the acute load. This method faces the same limitations as described above. Measuring injury risk for the subsequent day avoids the bias where low ACWRs are observed for injured athletes, but ignores any activity on that day that would have contributed to injury. On the other hand, including the current day's activity in the acute load calculation will result in lower ACWRs for athletes who get injured early in the day, albeit with reduced bias compared to the weekly calculations.

The underlying issue in the established methods is that injured and non-injured athletes are not matched on time. This creates a bias because athletes who get injured contribute less time during which they can add to their load. This limitation extends to any analysis where exposure time differs between injured and non-injured athletes and must be resolved to generate valid training recommendations.

3.4 The ACWR-Injury Risk Model Does Not Address Unmeasured and Time-Varying Confounding

For most causal questions concerning activity and injury, there likely exist important confounding factors that were unmeasured or not measured well. For instance, Bornn et al. used simulations to demonstrate that training schedule, which has seldom been adjusted for in studies on load and injury occurrence, may be a confounder and induce associations between the ACWR and injury even when there are none [44]. Novel strategies must be able to address unmeasured confounding to create generalizable recommendations.

A further challenge with training schedule and other potential confounders is that they are time-varying measures [45] that are affected by other time-dependent variables. For instance, loads may cause athletes to develop aches and pains that, while not considered injuries, may both predispose them to injury (affecting the outcome) and change their load in the following week (affecting the exposure). Because loads affect time-dependent variables that in turn affect the outcome, both time-varying confounding and time-varying confounding by past exposure (load) exist [46]. Traditional logistic regression methods as have been used with the ACWR, as well as time-dependent Cox proportional hazard models, random-effects models, and generalised estimating equations, are not appropriate for addressing time-varying confounding by past exposure [45]. Instead, advanced causal inference-based methods such as g-methods [46, 47] and other techniques [48] should be incorporated. To properly identify and model potential confounders, we suggest researchers construct a causal model and use it to inform

analyses [49]. We illustrate a possible directed acyclic graph for activity and injury in Sect. 4.1.

3.5 Difficulties Applying the ACWR to Subsequent Injuries

The ACWR has been used to develop training recommendations to prevent injuries. In fact, many injuries occur in athletes who were previously injured. These subsequent injuries make up a large portion of injuries [50–52], and as such, using an activity-injury risk model to guide their prevention may be of interest.

There are several considerations that are unique to subsequent injuries and were ignored in the ACWR-injury risk model. First, research using the ACWR has mainly used conventional logistic regression, which treats injuries in each athlete as independent events. An injury that occurs in each of 4 athletes should be analyzed differently than 4 injuries that occur in one athlete, as the latter is a repeated measures outcome. Second, injuries lead to changes in training schedules, as athletes often lose training time upon getting injured [53]. Third, injuries lead to changes in load capacity. Subsequent injuries may occur because athletes are not completely rehabilitated from injury. Training without having completely recovered from injury may put them at higher risk for injury than a healthy athlete, and they may experience an injury at lower loads because of the previous under-rehabilitated injury rather than the load. Subsequent injury models need to include time-varying confounders and approaches such as g-methods [47] will likely be required.

4 Alternative Strategies to Quantify the Effect of Activity On Injury

In this section, we propose alternative strategies to determine the effect of physical activity on injury risk that minimize the strong assumptions and limitations associated with the ACWR and how it has been implemented. These strategies avoid using a simple ratio measure to represent changes in activity and attempt to address limitations common in studies on activity and injury occurrence. Following our discussion in Sect. 2, any such strategy should aim to use continuous data. Although some authors have proposed restricting analyses to acute loads [23], this assumes that previous activity (i.e. chronic load) does not affect the capacity of tissues to withstand increases in load. We prefer not to make this assumption.

4.1 A Causal Framework for Activity and Injury

Causal inference approaches may be useful in analyzing how activity affects injury risk while accounting for time-dependencies of activity and confounders such as training schedule. We present a directed acyclic graph that illustrates a possible relationship between load, injury, and their confounders (Fig. 5). Instead of using one ratio, each load is its own independent variable that can cause injury independently or in combination with other loads. There may also be common causes of loads and/or injury such as training schedule. Furthermore, an injury may cause changes in subsequent loads, which may cause a subsequent injury.

Several relevant questions now arise. We may ask "How does load in week (or day) t_{k-1} affect injury risk in week (or day) t_k", "How does load in week t_{k-2} affect the relationship between activity in week t_{k-1} and injury in week t_k", "Does load in week t_{k-2} affect injury independent of activity in week t_{k-1}", "How does load in week t_{k-3} affect the relationship between activity in weeks t_{k-2} and t_{k-1} and injury in week t_k", and so forth. These are mediation-interaction questions that can be answered using their appropriate analyses [54]. Like the ACWR, these analyses take into account current and past activity, but without the assumptions imposed by ratios or only using acute load.

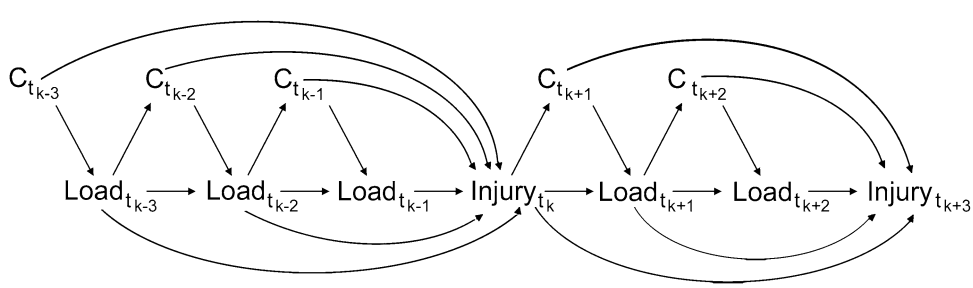

Fig. 5 Theoretical causal directed acylic graph showing the relationship between load and injury over time (t). Injury at time k (t_k) is caused by loads at t_{k-1}, t_{k-2}, t_{k-3}, or a combination of them. There are also common causes (C) of loads and/or injury. Injury also affects subsequent loads, which may cause subsequent injury

4.2 Using Instrumental Variables to Address Unmeasured Confounding

When using data to infer causation, it is critical to minimize unmeasured confounding. One strategy for obtaining an unbiased causal estimate of exposure on outcome in the presence of unmeasured confounding is to use an instrumental variable (IV) approach [55], provided IV assumptions hold [56]. An instrument is a variable that causes the exposure, does not cause the outcome except through its effects on exposure, and is not caused by any confounders of the exposure and outcome relationship.

In the context of activity and injury, an athlete's proposed training schedule might be a valid instrument for their actual activity because it would only affect injury through the actual activity performed. The instrument could be a binary variable (e.g. followed vs. did not follow the proposed training schedule), or continuous (e.g. proposed minutes of training each day). If a coach is aware of factors that increase the risk of injury for one of their athletes, they might modify the training schedule for that athlete. If measured, these confounding factors can be conditioned on in the IV analysis to satisfy the assumption that there is no confounding between the IV and the outcome. When collecting activity and injury data, we recommend that investigators record any reasons for changes in training schedules that might be related to injury risk to allow for more advanced causal inference type analyses.

4.3 Remaining Issues: Matching Injured and Non-injured Athletes on Time

Although these strategies minimize the major limitations associated with the ACWR and how it has been implemented, there remains an issue of bias due to differences in exposure time between injured and non-injured athletes. Injured and non-injured athletes can be matched on time in a nested case–control or case-crossover design. In a nested case–control design, one might match athletes based on exposure time (e.g. athletes sharing a training schedule), and "censor" the uninjured athletes at the calendar time that their matched athlete was injured. Therefore, all activity performed by both athletes prior to the injury is included, and activity performed after the injury excluded. In a case-crossover design, one would restrict the study population to athletes who had an injury [57] and compare their activity over a period of time just before they got injured with their activity at another period in time where they did not get injured. Both designs would reduce bias from differences in exposure time when modelling activity and injury data. This would lead to more appropriate estimates of the effect of activity on injury risk. However, both the matched nested case–control and case-crossover study designs censor data from the uninjured athletes, discarding useful information. They also suffer from limitations in interpretation. The case-crossover design only informs the relationship between activity and injury in those who experience injury over the time period of observation. The nested-case control design only informs the relationship in athletes with injuries compared to athletes who could have been matched to the injured athletes. They do not necessarily provide information on the average causal relationship between activity and injury in the entire population, without additional assumptions.

Although matching on time provides a solution when activity is recorded daily, many studies only gather weekly activity data for feasibility reasons. In these studies, an unbiased estimate of load could be estimated with some reasonable assumptions if the "training schedule" is known. To illustrate, consider two athletes that have similar training schedules. Athlete 1 plans for 2 h of activity each day for the current week, but gets injured on Tuesday at the end of the workout, totalling 4 h of activity. Athlete 2 plans for 1 h of activity each day, and completes the week without injury, totalling 7 h of activity. In this example, Athlete 1's load is 2/7 of their weekly load ($2/7 \times 14$ h $= 4$ h). Although Athlete 2 completed 100% of their load during the week, we could censor their activity at 2/7 of their weekly load ($2/7 \times 7$ h $= 2$ h). This simple calculation will work under the strong assumption that daily activity is equal throughout the week. Alternatively, if the planned training schedule is known and generally followed, one could use the planned load as a proxy for true load. We strongly recommend that training schedules be recorded whenever possible, and used as a proxy for actual activity where reasonable.

5 Conclusion

The ACWR has commonly been used to predict the relationship between activity and injury risk and interpreted causally to generate training recommendations. However, it suffers from several serious limitations. As a simple ratio, the ACWR may not be the most meaningful measure of changes in load. The EWMA method gives loads further in the past less weight but creates an "initial load" problem. The established ACWR-injury risk model was developed by treating continuous data as discretized in the original analyses, discretized data as continuous in evidence synthesis, and suffers from sparse data bias. When addressing the causal relationship between changes in load and injury risk, there are additional limitations such as unmeasured confounding, time-dependent confounding, and recurrent injuries. New analytical strategies such as causal inference approaches may address many of these limitations and should be explored as alternatives to currently used methods.

Compliance with Ethical Standards

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Conflict of Interest Chinchin Wang, Jorge Trejo Vargas, Tyrel Stokes, Russell Steele, and Ian Shrier declare that they have no conflicts of interest relevant to the content of this review.

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